

The second phase of my work at LISSI explored a hybrid reinforcement-learning framework for enabling the humanoid robot Pepper to autonomously navigate toward targets and track them in dynamic, human-populated environments. Rather than optimizing navigation performance alone, our aim was to develop an embodied agent capable of perceiving human behavior and adapting its actions to social conventions. To this end, we combined Proximal Policy Optimization (PPO) with Bayesian Neural Networks (BNNs) and Counterfactual Trajectory Active Learning (CTAL).

To equip the agent with richer representations of human motion, we pursued two complementary perception approaches. First, we built a human-motion dataset consisting of synchronized video recordings and on-body IMU measurements collected from participants performing varied movements, postures, and walking patterns. A TimeSformer coupled with a regression head was trained to infer quantitative motion information, including acceleration and angular velocity, directly from video. This enabled the robot to estimate human motion dynamics without requiring people to wear sensors during deployment. Second, we developed a MotionBERT-based pipeline that transformed visual 2D human poses into temporally consistent 3D body-joint representations, capturing the evolution of human movement over time.

The resulting predicted IMU dynamics and 4D human-motion representations, together with the robot's environmental and task-related information, were integrated into its state representation and processed through a Graph Neural Network (GNN). The PPO policy could therefore condition its decisions not only on the navigation context but also on the observed and anticipated motion of nearby humans. Finally, we introduced a social reward mechanism that compared the robot's generated behavior with positive and negative examples of human-robot interactions, rewarding similarity to socially appropriate behaviors and penalizing undesirable ones. Combined with conventional navigation rewards, this framework encouraged Pepper to simultaneously achieve effective, safe, and socially aware navigation.

Phase 2 is currently under submission to the International Conference on Robotics, Automation, and Artificial Intelligence 2026 (RAAI) that will be organized in Singapore

RESULTS

Five reinforcement learning approaches were compared: NNN-PPO (standard PPO), BNN-PPO (PPO with Bayesian Neural Networks), MCD-PPO (PPO with Monte Carlo Dropout), BNN-CTAL-PPO (Bayesian PPO enhanced with Counterfactual Trajectory Active Learning), and MCD-CTAL-PPO (Monte Carlo Dropout PPO enhanced with CTAL). These methods were benchmarked under identical training and testing conditions to assess the contribution of uncertainty estimation and active learning to autonomous robot navigation.

Best performances were obtained with BNN-CTAL-PPO

TRAINING RESULTS (64 Targets)

Metric	NNN-PPO	BNN-PPO	MCD-PPO	BNN-CTAL-PPO	MCD-CTAL-PPO
Average Cumulative Reward ↑	164	198	189	254	239
Episodes to Convergence ↓	4800	3600	3900	2500	3000
Training Time (days) ↓	2.2	2.7	2.5	3.1	2.9

TEST RESULTS (80 Unseen Targets)

Metric	NNN-PPO	BNN-PPO	MCD-PPO	BNN-CTAL-PPO	MCD-CTAL-PPO
Navigation Success Rate (%) ↑	52.5	63.75	60.0	81.25	73.75
Timeout Rate (%) ↓	26.25	21.25	21.25	10.0	13.75
Collision Rate (%) ↓	21.25	15.0	18.75	8.75	12.5
Average Navigation Time (s) ↓	74.3	61.5	65.8	45.2	50.1
Average Trajectory Length (m) ↓	18.7	16.2	16.9	13.8	14.6